

Self-Supervised Latent Topology Extraction of Physical Fields with Multiscale-JEPA

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Scientific discovery often begins by finding a representation in which hidden patterns become organized and interpretable. For continuous physical fields, such representations are difficult to define: classical statistics compress away spatial organization, while structure-aware methods require hand-crafted definitions of filaments, vortices, shocks, or voids. We introduce **Multiscale-JEPA**, a self-supervised framework that learns dense latent coordinates directly from scalar fields. Constrained Diffusion Decomposition (CDD; ?) acts as a physics-informed frontend that exposes pixel-registered, scale-separated morphological components, replacing raw intensity with structured physical primitives. A scale-aware Joint Embedding Predictive Architecture (??) is then trained to predict hidden target representations from visible context, [with mask size coupled to diffusion scale, aligning the predictive task with the physical scale hierarchy and ensuring that the structures the model must infer determine the distinctions its embedding preserves](#). Applied to MHD turbulence, nighttime-light maps, and PHANGS-ALMA molecular-gas observations, **Multiscale-JEPA** [organizes each field into coherent structural regimes: latent neighborhoods and extremal regions identify shock-like sheets, filamentary ridges, spiral-arm segments, and molecular-emission complexes without labels, hand-crafted segmentation rules, or domain-specific augmentations](#). The framework provides a label-free, task-adapted route to structural discovery in continuous physical fields.

INTRODUCTION

Scientific discovery often begins when a complex phenomenon can be seen in the right coordinates. Continuous physical fields have become central objects of investigation as attention shifts from isolated constituents toward patterns, structures, and collective organization. Examples range from dissipative vortex tubes in high-Reynolds-number turbulence (?) and magnetic reconnection layers in MHD flows (?) to the anisotropic matter distribution of the cosmic web (?). Existing methods typically take one of two routes: either they compress fields into global statistics such as spectra, probability distributions, and correlation functions, losing spatial and topological organization (???), or they identify structures with hand-crafted definitions, such as Hessian filaments (??), dendrogram clouds (?), and watershed voids (?). A methodological gap remains for methods that can organise the structural complexity of continuous physical fields without first compressing it away or prescribing in advance what the structures should be.

Latent understanding is central to both physical discovery and representation learning. In classical physics, the phase space of a system is well defined and its coordinates can often be prescribed by theory. For complex systems far from equilibrium, however, Kauffman identified this as a foundational difficulty (?): the relevant macroscopic variables—the phase space itself—cannot be prestatated but must be discovered. Traditional approaches such as Takens delay embeddings (?) and Koopman operator representations (?) offer principled constructions, but what one obtains in practice—independent delay coordinates or high-dimensional linearizations—is difficult to obtain and does

not extend naturally to spatially extended PDE fields with coherent structures nested across scales. The manifold hypothesis provides a geometric language for this gap: physical configurations occupy structured, low-dimensional subsets of a much larger ambient space (?). A useful latent representation should therefore act as a learned phase-space-like coordinate system for field morphology—one that preserves the organization of structures and their topological relations, rather than compressing them into aggregate statistics that are degenerate with respect to spatial arrangement and intermittency.

Self-supervised latent prediction offers a mechanism for learning such coordinates. Joint Embedding Predictive Architectures (JEPAs; ??) train a model to predict the latent representation of hidden regions from visible context, rather than reconstructing pixels. Conceived for perception tasks—autonomous driving, robotics, video understanding—JEPA’s world-model objective is equally well matched to physical science: pixel reconstruction spends capacity on high-frequency noise and irrelevant amplitude detail, while latent prediction emphasizes information that is predictable from surrounding physical context. Crucially, ? established that the masking strategy is the core design choice controlling what the model learns: target blocks must be large enough to force semantic inference rather than local interpolation, and context blocks must be spatially distributed enough to supply non-trivial structural information. Mask scale directly selects the level of the structural hierarchy the network is compelled to internalize.

We present **Multiscale-JEPA**, a self-supervised framework for learning dense latent coordinates of complex physical fields. The central difficulty is that physical structure is intrinsically multiscale: fine filaments, intermediate morphology, and large-scale envelopes are entangled in a single scalar image, while fixed-size masking

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probes only one arbitrary spatial scale. We address this by using Constrained Diffusion Decomposition (CDD; ?) in two complementary roles. First, CDD provides explicit scale-separated physical coordinates: a nonlinear diffusion decomposition that preserves sharp structural interfaces while reducing noise and cross-scale leakage. A scale-aware ConvNeXt encoder (?) processes these components before fusion, allowing fine, intermediate, and coarse structures to remain distinguishable in the learned representation. Second, CDD defines the masking geometry itself: the hidden footprint grows with diffusion scale, forcing the JEPA predictor to reason across the physical hierarchy rather than at a single fixed resolution. The resulting dense embeddings exhibit organized latent topology whose branches and extremal regions correspond to recognizable multiscale structures, enabling unsupervised structural discovery without hand-crafted labels, segmentation rules, or domain-specific augmentations.

I. METHOD: THE MULTISCALE-JEPA FRAMEWORK

A. Overview

We frame unsupervised representation learning for physical fields as a **Multiscale Joint Embedding Predictive Architecture (Multiscale-JEPA)**. The method uses a joint-embedding setup in which an on-line context branch receives a masked view of the field, an EMA target branch receives the unmasked field, and a lightweight predictor learns to infer target latent representations from context latent representations. Unlike reconstruction-based masked modeling, the objective is entirely latent: the model is not asked to reproduce raw pixels, but to preserve the information needed for context-to-target prediction.

The overall architecture is shown in Figure ??, and the CDD decomposition and scale-aware pyramid masking procedure are illustrated in Figure ??.

The predictive task is defined by the masking operator. For a context view x_c and target view x_t , masking specifies the conditional relation the model must learn: which aspects of the hidden target are predictable from the visible field. The JEPA embedding organizes the field according to this conditional structure, preserving variations that change the predictable target while compressing variations that do not affect the context-target relation (??). This perspective is consistent with the broader view in self-supervised learning that view construction determines which variations are treated as informative or invariant, and therefore shapes the geometry of the learned representation (???). In our setting, different masking regimes are therefore treated as different predictive tasks rather than as values of a single continuous hyperparameter.

Multiscale-JEPA implements this principle using a

physically structured view of the input. Constrained Diffusion Decomposition (CDD) decomposes the field into pixel-registered scale components, providing candidate physical scale coordinates. A scale-aware dense ConvNeXt encoder processes these components before fusion and produces a full-resolution latent map. We use fixed-box masking as a local spatial prediction baseline and CDD-scale-aware pyramid masking as the physically scale-aligned predictive task. A weak context spread regularizer prevents collapse during training. The resulting dense embedding is evaluated as a masking-induced organization of physical morphology: its significance lies in whether latent neighborhoods, branches, and extremal regions map back to coherent structures in the original field.

B. CDD Multiscale Coordinates

Standard encoders process physical fields as single scalar images, a choice that provides a weak inductive bias for multiscale phenomena such as turbulence. To address this, our framework processes a scale-ordered decomposition of the field generated with a Constrained Diffusion Decomposition (CDD) pyramid (?):

$$\frac{\partial I}{\partial t} = -\text{ReLU}(-\nabla^2 I) \quad (1)$$

Extracting components at exponentially spaced diffusion times yields a multiscale coordinate system $\mathcal{P}(x) = \{x^{(1)}, \dots, x^{(S)}\}$. CDD serves a vital physical and architectural role: it breaks the field down into scale-indexed morphological layers. This matters physically because it provides naturally grounded masking coordinates. By organizing the data across diffusion scales rather than arbitrary spatial patches, the architecture can explicitly separate macroscopic context from fine-scale structures, calibrating the predictive task to the actual physical hierarchy of the system.

C. Scale-Aware ConvNeXt Encoder

The CDD pyramid provides a set of pixel-registered scale components $\{x^{(1)}, \dots, x^{(S)}\}$, each representing structure at a different diffusion scale. We encode this pyramid with a scale-aware dense ConvNeXt backbone (?). Rather than concatenating all CDD channels at the input and treating them as an ordinary multichannel image, each scale is first processed through a per-scale adapter. The adapter maps the CDD component, its mask channel, and a scalar scale code into a common feature space. This allows fine-scale filaments, intermediate morphology, and coarse envelopes to form local features before competing in a fused representation.

Cross-scale fusion is deliberately delayed until after this per-scale adaptation step. This preserves scale identity during early feature extraction while still allowing

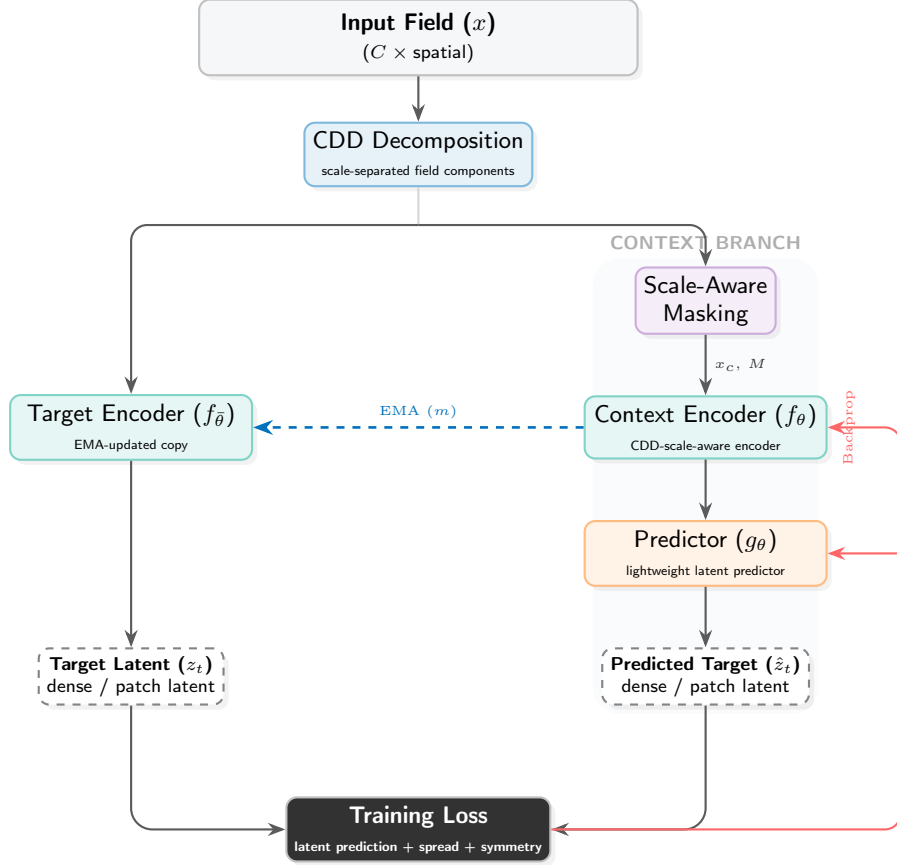


FIG. 1. **CDD-scale-aware JEPA architecture.** The CDD frontend decomposes the raw physical field into scale-separated components. The context branch applies scale-aware masking and encodes the masked context with a CDD-scale-aware encoder. The target branch bypasses masking and uses an EMA-updated copy of the encoder to provide target latent representations. A lightweight predictor maps the context representation to the target latent space. Training combines latent prediction, a weak context spread regularizer, and weak symmetry consistency. At inference, masking is disabled and the encoder produces dense latent maps for structural analysis.

masking_demo.pdf

FIG. 2. **CDD decomposition and pyramid masking.** *Left:* Constrained Diffusion Decomposition applied to a physical field, yielding a stack of dense, pixel-registered scale layers ordered from fine (top) to coarse (bottom). Each layer isolates coherent structures at a distinct physical scale — fine filaments and compact cores at small scales, extended morphology and large-scale envelopes at coarser scales — without spectral leakage between levels. *Right:* Scale-aware pyramid masking applied to the CDD stack. At each scale level, a spatially contiguous block is masked, with the masked region growing proportionally with scale. This forces the JEPA predictor to infer fine-scale structural targets from macroscopic context, calibrating the predictive bottleneck along the true physical hierarchy of the field rather than at an arbitrary fixed resolution.

the network to learn interactions between scales downstream. The fused multiscale feature map is then passed through a dense ConvNeXt encoder, which performs spatial mixing without reducing the field resolution. The

result is a full-resolution latent map

$$z = f_{\theta}(\mathcal{P}(x)), \quad (2)$$

where $\mathcal{P}(x)$ denotes the CDD pyramid. This dense latent

map is used by both the online context branch and the EMA target branch in the JEPA objective.

D. Multiscale Masking

To make the prediction task sensitive to the hierarchical structure exposed by CDD, we use **dynamic pyramid masking** as the primary task. Rather than applying a fixed spatial mask size across all channels, the context-mask footprint is tied to the CDD scale. For a CDD channel with scale factor S , the effective mask width is

$$B_{\text{eff}}(S) = \text{clip}(SB_{\text{base}} + B_{\text{offset}}, B_{\text{min}}, B_{\text{max}}), \quad (3)$$

where B_{min} sets the smallest nontrivial target size and B_{max} caps the footprint to remain within the encoder’s effective receptive field.

This pyramid strategy enlarges the hidden region for coarser CDD components and shrinks it for fine-scale components, forcing the predictive task to probe structures across the physical scale hierarchy rather than only at a fixed pixel scale. Masked regions are replaced by learned mask tokens specific to each CDD scale, explicitly signaling to the context encoder whether fine, intermediate, or coarse content is hidden. To prevent mask-induced artifacts, target locations are sampled with rejection constraints that avoid excessive footprint overlap. The implementation also allows manually defined per-channel mask sizes.

E. Training Objective and Latent-Space Analysis

The model follows a joint-embedding predictive architecture with an online context encoder f_θ , an EMA target encoder $f_{\bar{\theta}}$, and a lightweight spatial predictor g_θ . Given a masked context view x_c and an unmasked target view x_t , the model forms

$$z_c = f_\theta(x_c), \quad \hat{z}_t = g_\theta(z_c), \quad z_t = f_{\bar{\theta}}(x_t). \quad (4)$$

The target encoder is updated by exponential moving average,

$$\bar{\theta} \leftarrow m\bar{\theta} + (1 - m)\theta, \quad (5)$$

where m is the EMA momentum (??). The prediction loss compares predicted and target latent patches only at hidden target locations. For K sampled targets,

$$\mathcal{L}_{\text{pred}} = \frac{1}{K} \sum_{k=1}^K d(\hat{z}_{t,k}, z_{t,k}). \quad (6)$$

Here d is either the raw squared distance or its normalized variant. The raw form preserves amplitude information, while the normalized form compares latent directions and is useful when scale-balanced morphology is desired.

To prevent collapse, we apply a weak context spread regularizer to pooled valid context patches. Let $u_i \in \mathbb{R}^C$ be the pooled context embedding for valid target i . We compute the standard deviation σ_c of each channel across the batch and penalize only channels whose spread falls below a target value τ :

$$\mathcal{L}_{\text{spread}} = \frac{1}{C} \sum_{c=1}^C \max(0, \tau - \sigma_c). \quad (7)$$

This is a standard-deviation hinge: it provides an anti-collapse floor but does not impose whitening or a covariance penalty.

We also include a weak symmetry consistency loss for transformations T that preserve the physical meaning of the field,

$$\mathcal{L}_{\text{sym}} = \|\phi(f_\theta(Tx)) - \phi(Tf_\theta(x))\|_2^2, \quad (8)$$

where ϕ denotes the pooling or normalization used for comparison. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{spread}} \mathcal{L}_{\text{spread}} + \lambda_{\text{sym}} \mathcal{L}_{\text{sym}}. \quad (9)$$

At inference, masking is disabled and the encoder produces dense latent maps for the full field. These maps are used as label-free coordinates for inspecting learned representation geometry: neighborhoods, branches, and extremal latent regions can be mapped back to spatial locations to identify the physical structures represented by the network.

MULTISCALE-JEPA ON MHD TURBULENCE

We first evaluate Multiscale-JEPA on high-Reynolds-number simulations of compressible magnetohydrodynamic (MHD) turbulence. Such fields contain a mixture of diffuse regions, shear interfaces, filamentary ridges, and shock-compressed structures produced by the turbulent cascade. Our goal is not to reproduce classical diagnostics such as vorticity, current density, or density-gradient thresholds, but to test whether a self-supervised latent representation organizes these structures without explicit labels. The simulations follow the turbulent-cloud state-separation setting of ? and use open resources from the CATS portal (?).

Figure ?? shows the latent coordinates obtained from the frozen target encoder for the MHD turbulence field. A linear three-dimensional PCA projection provides a global view of the learned geometry, while UMAP emphasizes local neighborhood structure. Both projections reveal an organized latent manifold rather than a featureless cloud: dense regions, filamentary interfaces, diffuse void-like patches, and sharp transition zones occupy distinct but connected regions of latent space. We interpret these projections as exploratory maps of the masking-induced embedding, not as a unique physical phase diagram. Their value is that neighborhoods and extremal

jepa_mhd.pdf

FIG. 3. **Dense latent topology learned for MHD turbulence.** Target-encoder embeddings are projected to three dimensions using PCA and UMAP and mapped back to their original spatial locations. The PCA projection provides a linear view of the dominant latent directions, while the UMAP projection emphasizes nonlinear neighborhood structure in the learned representation. Both projections reveal spatially coherent organization: extended sheets, filamentary interfaces, and compact high-contrast structures occupy distinct regions of latent space. The inset point clouds show the corresponding three-dimensional projected latent distributions.

regions in latent space can be mapped back to spatial patches for physical inspection.

To examine the physical content of the embedding, we manually select representative regions of the latent manifold and map them back to the original MHD field. These

selections reveal several coherent structural regimes, including magnetically dominated diffuse cavities, kinetically dominated filamentary ridges, and sharp edge-like transition layers. The precise physical interpretation of these regions depends on auxiliary MHD variables and

jepa_ngc.pdf

FIG. 4. Dense latent topology learned for the NGC molecular-gas field. Target-encoder embeddings are projected with PCA and UMAP and back-mapped to the observed field. The PCA projection highlights large-scale latent variation across spiral-arm and interarm morphology, while the UMAP projection separates nonlinear structural neighborhoods associated with bright molecular complexes, diffuse emission, and elongated arm-like features. The inset point clouds show the projected latent geometry, indicating that the representation forms a continuous but structured manifold rather than a set of discrete supervised classes.

will be refined with additional diagnostics, but the key result is that the latent geometry separates recognizable structures without using those variables during training.

The manually selected latent regions provide a qualitative check on the latent map shown in Figure ??: patches

drawn from nearby latent neighborhoods exhibit related morphology in the original field, while movement across the manifold corresponds to systematic changes such as diffuse-to-filamentary transitions and edge-to-sheet-like geometries. This suggests that the dense embedding is

jepa_chengdu.pdf

FIG. 5. **Dense latent topology learned for the Chengdu nighttime-light field.** Target-encoder embeddings are projected with PCA and UMAP and mapped back to their original spatial locations. The projections reveal coherent latent organization across the urban field, separating compact high-intensity cores, extended road-like structures, diffuse emission, and low-intensity surrounding regions. The inset shows the original nighttime-light image, providing the spatial reference for the latent back-mapping.

not merely encoding raw intensity, but provides a useful coordinate system for exploring coherent structures in the turbulent field.

If these manually selected MHD patch examples are retained as a main result, insert a dedicated mapped-back patch figure here; otherwise this paragraph can remain

as a qualitative description of the inspection procedure.

CROSS-DOMAIN APPLICATION EXAMPLES

Figures ?? and ?? show two cross-domain applications of the same Multiscale-JEPA framework: nighttime lights in Chengdu and molecular-gas emission in a nearby galaxy. These examples are not intended as exhaustive domain-specific studies. Instead, they test whether the learned dense embeddings remain interpretable when the input changes from simulated turbulence to urban and extragalactic scalar fields.

a. Nighttime lights in Chengdu. We apply Multiscale-JEPA to a nighttime-light map of Chengdu as a non-astrophysical scalar-field test (Figure ??). The field is shaped by urban light distribution rather than by fluid or gravitational dynamics, providing a useful contrast to the simulated and astronomical fields considered elsewhere in this work. The embedding separates compact bright urban regions, elongated light corridors, patchy suburban structure, and low-intensity background into distinct latent neighborhoods. We interpret this result only as structural organization of the light field, not as a supervised land-use or socioeconomic classification.

b. Molecular gas in NGC 3927. We also apply Multiscale-JEPA to the $^{12}\text{CO}(J = 2-1)$ integrated-intensity map of NGC 3927 from the PHANGS-ALMA survey (?) (Figure ??). In this case, the learned embedding separates molecular-gas structures by their spatial organization. Strongly concentrated regions, including the galaxy center and dense arm segments, occupy distinct latent neighborhoods from more diffuse spiral-arm and interarm gas. The separation is driven by structural complexity rather than by labels: compact peaks, arm-like ridges, and lower-contrast interarm emission appear as different regions of the latent topology.

Together, these examples show that Multiscale-JEPA can be used as a label-free exploratory tool across domains. The same training recipe produces dense embeddings whose neighborhoods correspond to coherent spatial structures, while the interpretation of each latent region remains domain-specific.

CONCLUSION

How can we organize complex physical fields when the relevant structures are not known in advance? This work approached that question through **Multiscale-JEPA**, a self-supervised framework that learns dense latent coordinates by predicting hidden target representations from visible context. Rather than reconstructing pixels or imposing hand-crafted structure labels, the model learns the distinctions that are useful for a chosen context-to-target prediction task.

The central ingredient is the combination of physically grounded multiscale coordinates with latent predictive learning. Constrained Diffusion Decomposition (CDD) exposes pixel-registered scale components of the input

field, while a scale-aware encoder preserves these components before fusion. The JEPA objective then organizes the representation according to the conditional structure induced by masking: variations that affect prediction are preserved, while variations irrelevant to the context-target relation are compressed.

Across MHD turbulence, nighttime-light structure, and molecular-gas emission, the resulting embeddings form organized latent topologies whose neighborhoods and extremal regions map back to coherent spatial structures. These examples show that Multiscale-JEPA can serve as a label-free exploratory tool for multiscale scalar fields, revealing structural regimes without explicit segmentation rules or auxiliary physical diagnostics.

The experiments expose the link between masking and representation. Fixed-box, randomized, and CDD-scale-aware masks do not merely change training difficulty; they ask different predictive questions and open different latent organizations of the same field. This makes Multiscale-JEPA a useful test ground for studying how masking defines the geometry of self-supervised representations in continuous physical data. A fuller account of this relationship—including scale-energy balancing, semantic-scale selection, and morphology-aware target design—remains open, but the present framework provides the machinery needed to investigate it.

Multiscale-JEPA opens a practical route for predictive representation learning in physical fields, where the relevant coordinates are often unknown and the structures of interest emerge only after the data are reorganized. By constructing dense latent embeddings that can be inspected, segmented, and mapped back to the original field, the framework enables automatic extraction of latent structure without requiring the user to specify filaments, voids, shocks, arms, or clouds in advance. It therefore provides a rigorous foundation for structural segmentation and discovery: one that can validate known physical regimes while also exposing new candidate patterns before a complete vocabulary for them exists.

Appendix A: CDD: Numerical Implementation

Scale Extraction and Normalization

The CDD process evolves the field I according to

$$\frac{\partial I}{\partial t} = \min(0, \nabla^2 I),$$

where ∇^2 is computed with a five-point stencil and reflective boundary conditions. We sample diffusion times

$$\{t_k\}_{k=1}^N = \{1, 2, 4, 8, 16, 32\}, \quad N = 6.$$

Scale primitives are defined as consecutive differences for $k = 1, \dots, N - 1$:

$$P_k = I^{t_k} - I^{t_{k+1}}, \quad (\text{A1})$$

and the coarsest residual channel is

$$P_N = I^{t_N}. \quad (\text{A2})$$

Thus, the representation has N channels in total. Each channel captures structure concentrated in a diffusion scale band, with characteristic length $\lambda_k \propto \sqrt{t_k}$.

To handle high dynamic range, each channel is normalized independently using percentile clipping followed by shifted log compression. Let $q_k^{(1)}$ and $q_k^{(99)}$ denote the 1st and 99th percentiles for channel k , estimated on the training split and fixed for validation/test. Define

$$P'_k = \text{clip}(P_k, q_k^{(1)}, q_k^{(99)}), \quad (\text{A3})$$

$$\hat{P}_k = \frac{P'_k - q_k^{(1)}}{q_k^{(99)} - q_k^{(1)} + \varepsilon} \in [0, 1], \quad (\text{A4})$$

$$\tilde{P}_k = \frac{\log(1 + \alpha \hat{P}_k)}{\log(1 + \alpha)}, \quad \tilde{P}_k \in [0, 1], \quad (\text{A5})$$

where $\varepsilon > 0$ prevents division by zero and $\alpha > 0$ controls compression strength (larger α gives stronger compression near high amplitudes). This mapping suppresses outliers, stabilises dynamic range, and preserves relative ordering of sub-dominant structures within each scale channel.

Scale-Aware Encoder Implementation

Figure ?? summarizes the implementation of the scale-aware ConvNeXt encoder used in Multiscale-JEPA. The CDD decomposition is treated as a separate preprocessing step that produces a pixel-registered pyramid of scale components

$$\mathcal{P}(x) = \{x^{(1)}, \dots, x^{(S)}\}.$$

Unlike a conventional feature pyramid, these levels are not produced by progressive downsampling inside the neural network. They are physical scale components obtained before learning, and all remain registered at the input resolution.

The encoder begins after this decomposition. For each scale s , the encoder forms a per-scale input by stacking the CDD component, the corresponding mask channel, and a scalar scale code. A shared scale-adaptor architecture maps this input into a common feature width,

$$h^{(s)} = A_\theta(x^{(s)}, m^{(s)}, c^{(s)}), \quad s = 1, \dots, S,$$

where $m^{(s)}$ is the mask channel and $c^{(s)}$ is the scale code. The adaptor therefore does not see the CDD value alone; it also receives information about which pixels were hidden and which physical diffusion scale the channel represents. This gives each scale local spatial processing before cross-scale fusion and places all CDD components into a comparable feature space.

The adapted features are fused across scale,

$$h = \text{Fuse}(h^{(1)}, \dots, h^{(S)}).$$

In the final configuration, Fuse is implemented as a top-down coarse-to-fine fusion: coarser-scale features are progressively injected into finer-scale features before the fused representation is passed to the ConvNeXt blocks. This differs from a standard feature pyramid because the hierarchy is supplied by CDD before learning rather than generated by neural downsampling. The fused multiscale feature map is then processed by a dense ConvNeXt backbone,

$$z = F_\theta(h),$$

which performs spatial mixing without reducing the field resolution. The same encoder architecture is used for the online context branch and for the EMA target branch. Default implementation details are listed in Table ??.

Several normalization layers are used to make the CDD channels comparable before fusion. Per-scale adaptor normalization is applied separately within each scale branch, preventing high-amplitude CDD channels from dominating the shared adaptor. Adaptor normalization stabilizes the shared convolutional frontend, while stem normalization stabilizes the first ConvNeXt projection after scale fusion. A final LayerNorm is applied after the encoder head to place the dense latent vectors on a common scale before prediction and analysis. These normalization choices are summarized in Table ??.

Projector and Predictor

The encoder outputs a dense latent map

$$z_c \in \mathbb{R}^{C \times H \times W}.$$

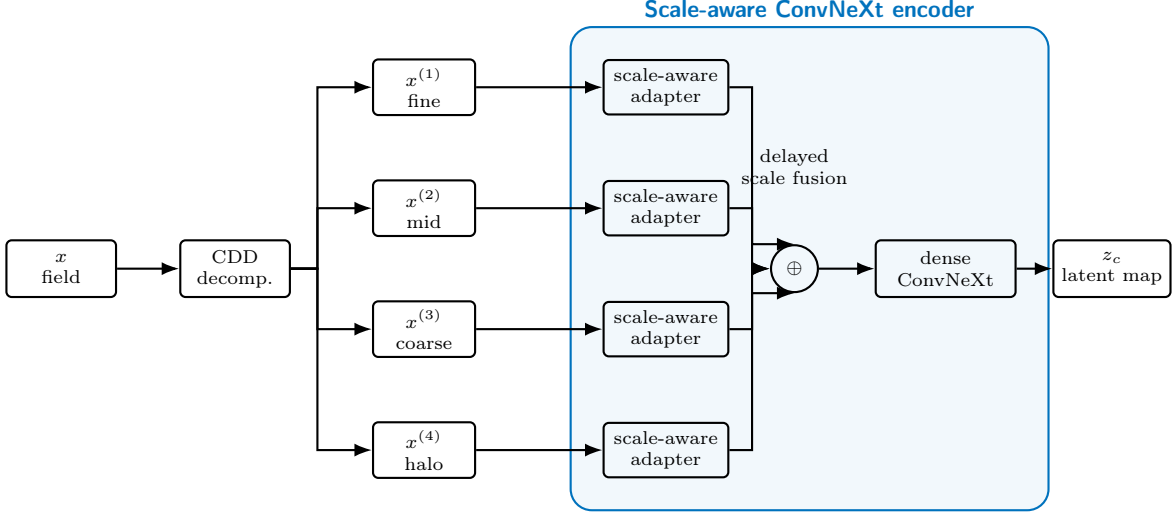


FIG. 6. **Scale-aware ConvNeXt encoder.** The input field x is first decomposed into a CDD pyramid of pixel-registered scale components. The encoder then processes each CDD component along a per-scale adapter path before cross-scale fusion, preserving scale identity before spatial mixing. The fused multiscale representation is passed through a dense ConvNeXt encoder to produce the full-resolution latent map z_c .

TABLE I. **Default scale-aware ConvNeXt encoder settings.** Implementation defaults used in the reported experiments, grouped by functional role.

Component	Default setting
<i>Input and scale representation</i>	
Input representation	CDD pyramid
CDD scales	[2, 4, 8, 16]
Per-scale adapter input	field, mask channel, scale code
<i>Per-scale adaptation</i>	
Adapter sharing	shared architecture across scales
Adapter kernel	3×3
Adapter feature width	8 channels per scale
<i>Cross-scale fusion</i>	
Scale fusion	top-down coarse-to-fine fusion
Spatial downsampling	none
<i>Dense ConvNeXt backbone</i>	
Encoder backbone	dense ConvNeXt
Encoder width	64 channels
Encoder depth	4 blocks
ConvNeXt spatial kernel	7×7 depthwise convolution
Global response normalization	enabled
<i>Latent output</i>	
Latent channels	32
Final normalization	LayerNorm
Output	dense latent map $z \in \mathbb{R}^{32 \times H \times W}$

Before prediction, we optionally apply a pointwise projector that refines the latent channels without changing the spatial resolution,

$$q_c = p_\theta(z_c).$$

The projector is implemented as a lightweight 1×1 convolutional bottleneck and preserves the latent dimensionality.

The predictor maps the context representation to the target embedding space using a lightweight full-resolution convolutional head,

$$\hat{z}_t = g_\theta(q_c).$$

In the final configuration, g_θ includes local spatial convolution, so target predictions can depend on neighboring context features rather than only on pointwise channel mixing. Predicted patches are extracted at hidden target locations and compared with the corresponding EMA target-encoder patches in the JEPA loss.

No global pooling is applied: the predictor remains dense, and the output retains the spatial resolution of the input field. This is essential because the learned representation is used as a pixel-level latent map for structural analysis.

Mixed precision

Training uses PyTorch Automatic Mixed Precision (AMP) (?) with gradient scaling. On supported hardware, we also enable TF32 to improve throughput.

Appendix B: Training Configuration

This section records the default training configuration used for the final experiments. Sweep results and run selection are reported separately in Appendix ???. The

purpose here is to define the data preprocessing, masking parameterization, target sampling, loss terms, and optimization recipe.

Data and Preprocessing

The input file pattern is dataset-specific. For the MHD experiments, we use the density slice from the selected simulation run. All datasets are treated as scalar fields and converted to the CDD pyramid before training. CDD is run in constrained logarithmic mode, and the same post-log transform is applied to all channels to stabilize high-dynamic-range structure. This produces pixel-registered scale components that serve as the input to the scale-aware encoder.

During training, we apply D4 augmentations, consisting of rotations and flips that preserve the scalar-field structure. During inference, masking is disabled and the frozen target encoder is applied to the full field. Flip-based test-time augmentation is enabled for the final dense latent maps.

Mask Parameterization and Target Sampling

Masking defines the context–target prediction task. For each CDD scale with diffusion scale σ_s , the effective target footprint is computed as

$$B_s = \sigma_s f_{\text{mask}} + B_0, \quad (\text{B1})$$

where f_{mask} is the scale multiplier and B_0 is the fixed footprint in pixels. Setting $f_{\text{mask}} = 0$ gives a fixed-box mask, while setting $B_0 = 0$ gives a pure scale-tied pyramid mask. Randomized pyramid runs sample f_{mask} from a specified interval during training.

Candidate target centers are generated on a grid whose spacing is proportional to the effective footprint,

$$D_s = \gamma_{\text{mask}} B_s, \quad (\text{B2})$$

with $\gamma_{\text{mask}} = 2.0$ in the final configuration. This spacing limits excessive redundancy while still allowing target patches to overlap when `target_nonoverlap=false`.

Target selection can be either random or priority-based. In random mode, target centers are sampled uniformly from the candidate catalog. In priority mode, candidates are ranked by an image-derived score, such as flux, local intensity, or a CDD-scale contrast score, and sampling is restricted to the top fraction of this ranked catalog. The default configuration uses random target sampling unless otherwise stated. The masking sweep also includes priority-sampled candidates; the selected MHD run uses priority sampling, while the selected Chengdu and NGC runs use random sampling.

TABLE II. **Masking and target-sampling defaults.** The masking parameters define the context–target prediction task. Sweep results are reported separately.

Setting	Default / role
<code>mask_scale_factor</code>	scale-tied footprint multiplier f_{mask} ; swept in masking experiments
<code>mask_footprint_px</code>	fixed footprint B_0 in pixels; swept in masking experiments
<code>mask_spacing_scaling</code>	$\gamma_{\text{mask}} = 2.0$, grid spacing relative to footprint
<code>target_sampling_mode</code>	random by default; priority-sampled candidates included in sweeps
<code>priority_top_percent</code>	100 by default; lower values used for priority-sampled sweep candidates
<code>priority_n_target</code>	auto, image-area-based target count
<code>target_nonoverlap</code>	false, overlapping targets allowed

Loss Terms

The final training objective is dominated by the JEPA prediction loss. Predicted and target embeddings are compared only at masked target locations. We L2-normalize patch embeddings before the MSE loss, so the prediction objective emphasizes latent direction rather than raw feature magnitude.

A weak context spread regularizer prevents collapse by encouraging the standard deviation of pooled context embeddings to remain near one. A weak symmetric feature loss encourages consistency under four-way rotations. VICReg variance and covariance penalties are disabled in the final configuration.

Optimization Defaults

All final runs use AdamW with cosine learning-rate decay and EMA target-encoder updates. The predictor uses a spatial convolutional head so target predictions can depend on neighboring context features. LayerNorm inside the predictor is disabled in the final configuration.

Inference Diagnostics

For inference, masking is disabled and the frozen target encoder produces the dense latent map. Flip-based test-time augmentation is enabled. We compute effective-rank diagnostics as collapse and latent-use checks, and we enable the scale-response probe to measure how strongly the trained encoder depends on each CDD input channel.

TABLE III. Loss terms used in the final training configuration.

Term	Weight	Active	Role
Prediction MSE	50	yes	patch-level latent prediction loss
Spread regularizer	2	yes	std-hinge toward unit spread
Symmetric feature loss	0.003	yes	four-way rotational consistency

TABLE IV. Optimization defaults.

Setting	Default
<code>normalize_loss_l2</code>	enabled
<code>predictor_spatial_conv</code>	enabled
<code>predictor_hidden</code>	96
<code>predictor_layernorm</code>	disabled
Epochs	10
Batch size	2, 4
Optimizer	AdamW
Learning rate	10^{-4}
Minimum learning rate	10^{-6}
Warmup	1.0 epoch
Weight decay	10^{-5}
EMA momentum	$0.99 \rightarrow 0.9999$
EMA warmup fraction	0.25

TABLE V. Inference and diagnostic defaults.

Setting	Default
Inference TTA	enabled
TTA mode	<code>flip4</code>
Scale-response probe	enabled
Effective-rank diagnostic	enabled

Appendix C: Masking Sweeps and Run Selection

We used compact masking sweeps to select representative embeddings for the main visualizations. These sweeps were not intended to identify a universal optimal mask. Instead, they served as stability and interpretability checks. Each candidate run had to remain non-collapsed, use more than one latent direction, and produce a dense embedding whose neighborhoods could be mapped back to coherent spatial structures.

Across the final sweep, all runs remained healthy by collapse diagnostics: no dead latent channels were detected. Effective rank was used as a practical indicator of latent-space usage. Empirically, larger effective rank often corresponded to visually richer embeddings, but it was not used as the sole selection criterion. Final runs were selected by combining effective-rank diagnostics with visual inspection of PCA/UMAP maps and mapped-back spatial regions.

For a covariance spectrum with normalized eigenvalues p_i , effective rank is computed as $\exp[-\sum_i p_i \log(p_i)]$. The target participation number is computed from the same spectrum as $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$. The participation

ratio reports this participation relative to the available latent dimensionality. A dead channel is counted when its standard deviation over sampled spatial embeddings falls below the numerical activity threshold used in the diagnostic pass. These quantities are used only as collapse and latent-use diagnostics, not as standalone measures of physical quality.

The selected MHD embedding uses priority sampling with a pyramid mask at scale 1.2. Although some random-mask MHD runs reached higher effective rank, the priority-sampled run produced a cleaner and more interpretable latent map under visual inspection. This illustrates why effective rank is useful as a screening metric but not sufficient as the final selection criterion.

For Chengdu, the pyramid scale 1.2 run was selected because it produced the richest embedding in the final sweep and separated compact bright regions, extended light structures, and patchy lower-intensity areas more clearly than the compact random or priority alternatives.

For NGC, the full-sample pyramid scale 1.6 run was selected. It did not have the largest effective rank among all NGC candidates, but it gave the most stable dense map for the molecular-gas field and separated concentrated emission, arm-like structure, and lower-contrast interarm regions more cleanly than the smaller full-sample masks.

The selected-run diagnostics in Figure ?? show the input fields and training-loss histories for the three embeddings used in the main visualizations. In each case, the spread regularizer quickly becomes inactive after establishing a non-collapsed latent space, leaving the weighted prediction term as the dominant training signal. These loss histories provide optimization context for the selected dense latent maps.

Appendix D: Latent Map Visualization and Diagnostics

After training, masking is disabled and the frozen target encoder is applied to the full field to produce a dense latent map. For large fields, we evaluate overlapping windows and blend the outputs to reduce boundary artifacts.

For visualization, latent vectors are sampled from the dense map and projected with PCA or UMAP (?). PCA is used as a linear view of the global latent geometry, while UMAP is used as a nonlinear neighborhood-preserving view. The fitted projection is then evaluated over the full spatial map and normalized for RGB rendering.

TABLE VI. **Final Gen 125 sweep summary.** Effective rank is used as a latent-use diagnostic, while final selection also uses visual topology and physical interpretability. Selected runs are marked with an asterisk. All runs have zero dead channels.

Run	Dataset	Sampling	Masking	Eff. rank	Pred. rank	Target part.	Part. ratio
1	Chengdu	random	pyramid 0.8	1.798	1.566	1.39	0.965
2*	Chengdu	random	pyramid 1.2	3.174	1.496	1.99	0.601
3	Chengdu	random	pyramid 1.6	1.307	1.090	1.11	0.927
4	Chengdu	random	random pyramid 0.4–1.6	1.175	1.061	1.05	0.967
5	Chengdu	priority	pyramid 1.2	1.164	1.043	1.05	0.966
6	MHD	random	box 5	4.227	1.579	2.93	0.440
7	MHD	random	box 9	1.758	1.229	1.28	0.841
8	MHD	random	box 13	2.166	1.255	1.53	0.707
9	MHD	random	pyramid 0.4	1.844	1.424	1.35	0.899
10	MHD	random	pyramid 0.8	1.828	1.414	1.39	0.848
11	MHD	random	pyramid 1.2	1.533	1.192	1.20	0.891
12	MHD	random	pyramid 1.6	2.618	1.339	1.79	0.631
13	MHD	random	random box 3–15	3.145	1.260	1.97	0.553
14	MHD	random	random pyramid 0.4–1.6	2.323	1.731	1.65	0.844
15*	MHD	priority	pyramid 1.2	2.137	1.190	1.42	0.748
16	NGC	random	pyramid 0.8	1.201	1.099	1.06	0.968
17	NGC	random	pyramid 1.2	1.276	1.189	1.10	0.973
18	NGC	random	pyramid 1.6	1.362	1.124	1.13	0.924
19	NGC	random	random pyramid 0.4–1.6	1.979	1.484	1.53	0.794
20	NGC	random	pyramid 0.8, full sample	1.235	1.169	1.08	0.987
21	NGC	random	pyramid 1.2, full sample	1.185	1.056	1.06	0.959
22*	NGC	random	pyramid 1.6, full sample	1.757	1.178	1.31	0.813
23	NGC	random	random pyramid 0.4–1.6, full sample	1.252	1.154	1.09	0.967
24	NGC	priority	pyramid 1.2	1.288	1.080	1.10	0.935

* Selected run. Selection is based on visual topology and physical interpretability in addition to the latent-use diagnostics listed here.

We use effective rank and participation-style diagnostics only as collapse and latent-use checks. Low values indicate that the embedding has contracted to a small number of directions, while broader spectra indicate greater use of the latent space. These metrics are not treated as a standalone measure of physical quality; final interpretation comes from mapping latent neighborhoods and

extremal regions back to the original field.

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All experiments in this work were performed on self-funded consumer-grade hardware, primarily an Apple MacBook Pro with an M3 Pro chip and a headless desktop equipped with an NVIDIA RTX 3090 GPU.

